ULB UNIVERSITÉ LIBRE DE BRUXELLES

MLG

Python Crop Simulation Environment - PCSE

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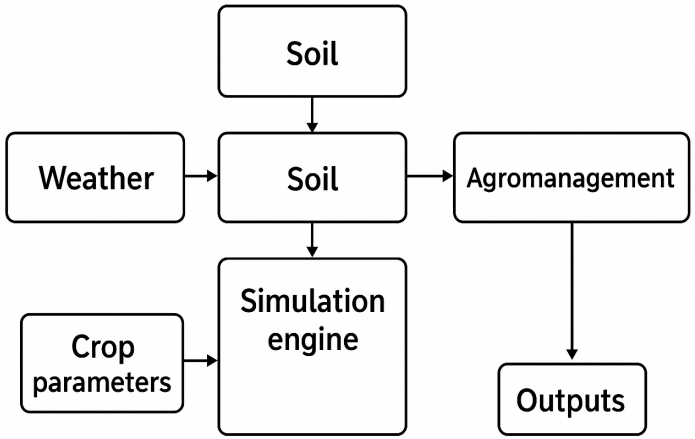
Motivation

- Agriculture is facing pressures from climate variability and sustainability constraints.
- **Digital twins** offer a virtual counterpart of the real agro-system for prediction, optimization, and decision support.
- Core requirement: a **process-based simulator** to generate synthetic but physically consistent data.
- We explore how the **Python Crop Simulation Environment (PCSE)** can play that role.

What is PCSE?

- **PCSE (Python Crop Simulation Environment)** - open-source framework from Wageningen University.
- Implements classical crop models such as:
  - **WOFOST** (World Food Studies model) - daily crop growth and yield.
  - **LINGRA** - grassland productivity.
- Written in Python, modular, composable, and easily coupled with external data or ML models.

Architecture Overview

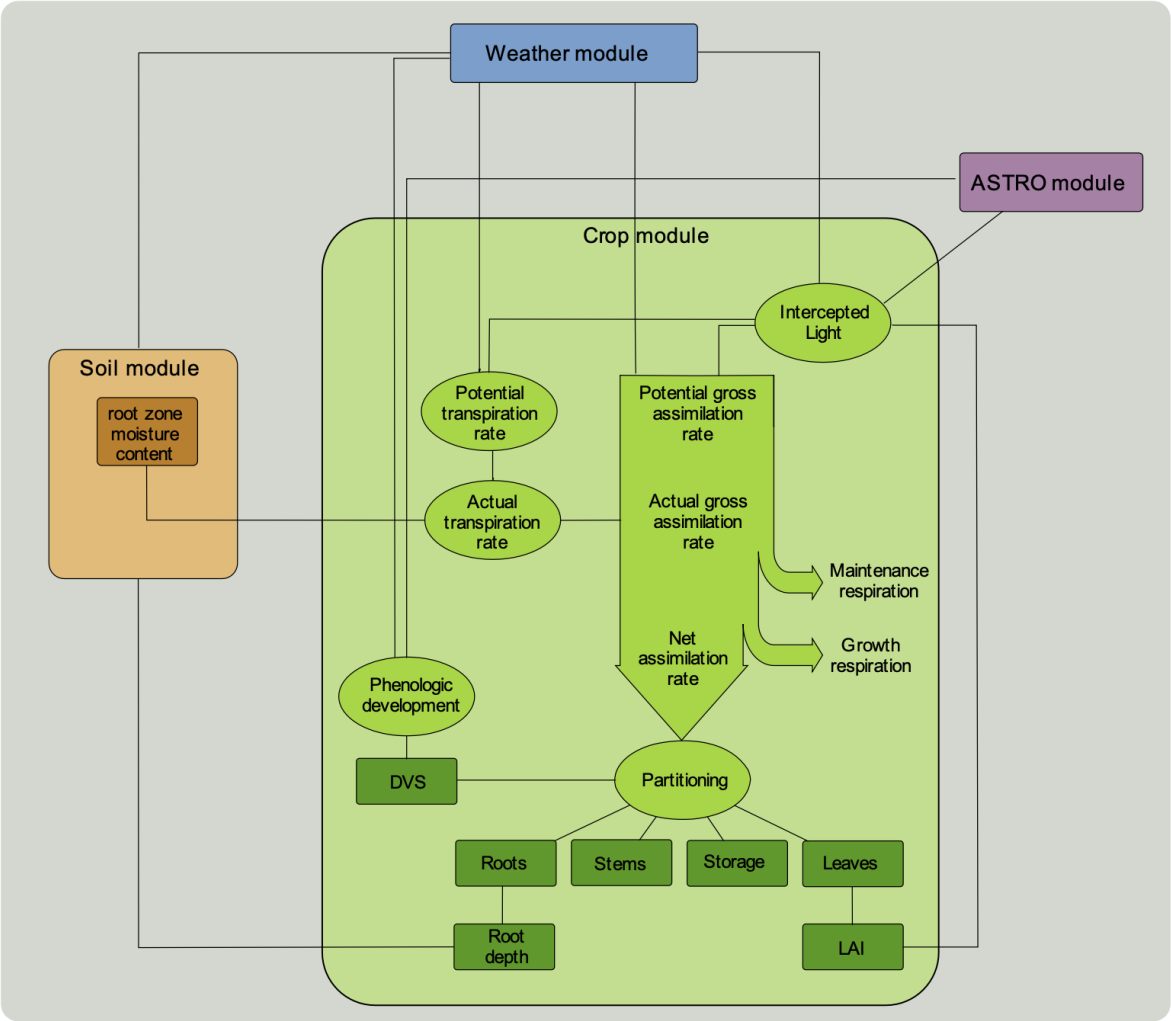


- Inputs:
  - Weather (NASA POWER, ECMWF, etc.)
  - Soil and crop parameters
  - Agromanagement schedule (sowing, irrigation)
- Simulation engine integrates crop ODEs daily.
- Outputs: biomass, LAI, soil water, yield, etc.

The WOFOST Model

- **WOFOST** (World Food Studies) is a mechanistic crop growth model developed by Wageningen University.
- It describes the daily development of a crop as a system of coupled ODEs:
  $\frac{dB_i}{dt} = f_i(\text{radiation, temperature, water, nutrients})$ where B_i are biomass components (leaves, stems, storage organs).
- **Core processes:**
  - Photosynthesis and respiration balance.
  - Partitioning of assimilates between organs.
  - Soil water balance and evapotranspiration.
  - Phenological development (temperature-driven).
- Two main modes (in version 7.2: last well documented one, but less complex):
  Potential production: optimal water and nutrients.
  Water-limited production: growth constrained by soil moisture.
- Provides interpretable outputs: LAI, biomass, yield, soil water, etc.

The WOFOST Model



Simplified general structure of the dynamic, explanatory crop growth model WOFOST

Phenological and Canopy Variables

- DVS - Development Stage
  - Dimensionless index defining crop phenological development.
  - Ranges from -0.1 (sowing) $\rightarrow$ 0.0 (emergence) $\rightarrow$ 1.0 (flowering) $\rightarrow$ 2.0 (physiological maturity).
  - DVS $= 1$: start of yield formation (grain filling, tuber bulking, etc.).
  - Nearly all model processes depend on DVS — calibration is critical.
- LAI - Leaf Area Index
  - One-sided green leaf area per unit ground area (dimensionless).
  - Controls light interception and photosynthesis.
  - When LAI $= 0$: canopy senesced, photosynthesis and transpiration stop.

Biomass Partitioning Variables

- **TAGP - Total Above-Ground Production:** total dry biomass (kg/ha). TAGP = TWSO + TWLV + TWST.
- **TWSO - Total Weight Storage Organs:** harvestable yield (grains, tubers, beans). Reported as dry weight; fresh yield depends on crop water content.
- **TWLV - Total Weight Leaves:** cumulative leaf biomass (living + dead). When LAI = 0, all leaf biomass is dead.
- **TWST - Total Weight Stems:** dry stem biomass, varies by crop species and cultivar.
- **TWRT - Total Weight Roots:** below-ground biomass. No explicit link to root depth in WOFOST.

Water and Soil Variables

- **TRA - Crop Transpiration (cm/day):** Water taken up from soil and transpired through leaves; zero when LAI=0.
- **RD - Rooting Depth (cm):** Minimum of maximum crop rootable depth and soil rootable depth.
- **SM - Soil Moisture (fraction):** Volumetric water content of the root zone ($0 =$ dry, $0.4 \approx$ saturated). Constant under potential production (non-limiting water).
- **WWLOW - Water in Rooted + Lower Zone:** Only used under water-limited simulations; undefined in potential runs.

Together, these variables describe the state of the crop-soil-water system simulated by WOFOST.

Example: Running WOFOST

```
cropdata = CABOFileReader('./sug0601.crop')
sitedata = WOFOST72SiteDataProvider(WAV=100)
soildata = CABOFileReader('./ec3.soil')
parameters = ParameterProvider(cropdata=cropdata,
    soildata=soildata, sitedata=sitedata)
weather = NASAPowerWeatherDataProvider(latitude=52.0,
    longitude=5.0)
agro = YAMLAgroManagementReader("./sugarbeet_calendar.agro")
wofost = Wofost72_WLP_CWB(parameters, weather, agro)
wofost.run_till_terminate()
results = wofost.get_output()
```

Outputs: daily time series of LAI, biomass, evapotranspiration, yield, etc.

PCSE as a Data Generator

- Generates structured, physically interpretable datasets:

$$\{x_t = (\text{weather, soil, crop}),\ y_t = (\text{LAI, biomass, yield, ...})\}$$

- By varying:
  - Geographic location and weather,
  - Soil/crop types (many crops descriptions files are available on *Github*, soil descriptions in *this website*),
  - Management practices (listed in the *WOFOST website*),

  we can obtain a rich synthetic dataset
- Applications:
  - ML pretraining or surrogate modeling,
  - Data assimilation and state estimation,
  - Scenario analysis and control policy testing.
  - **Data generation for a digital twin**

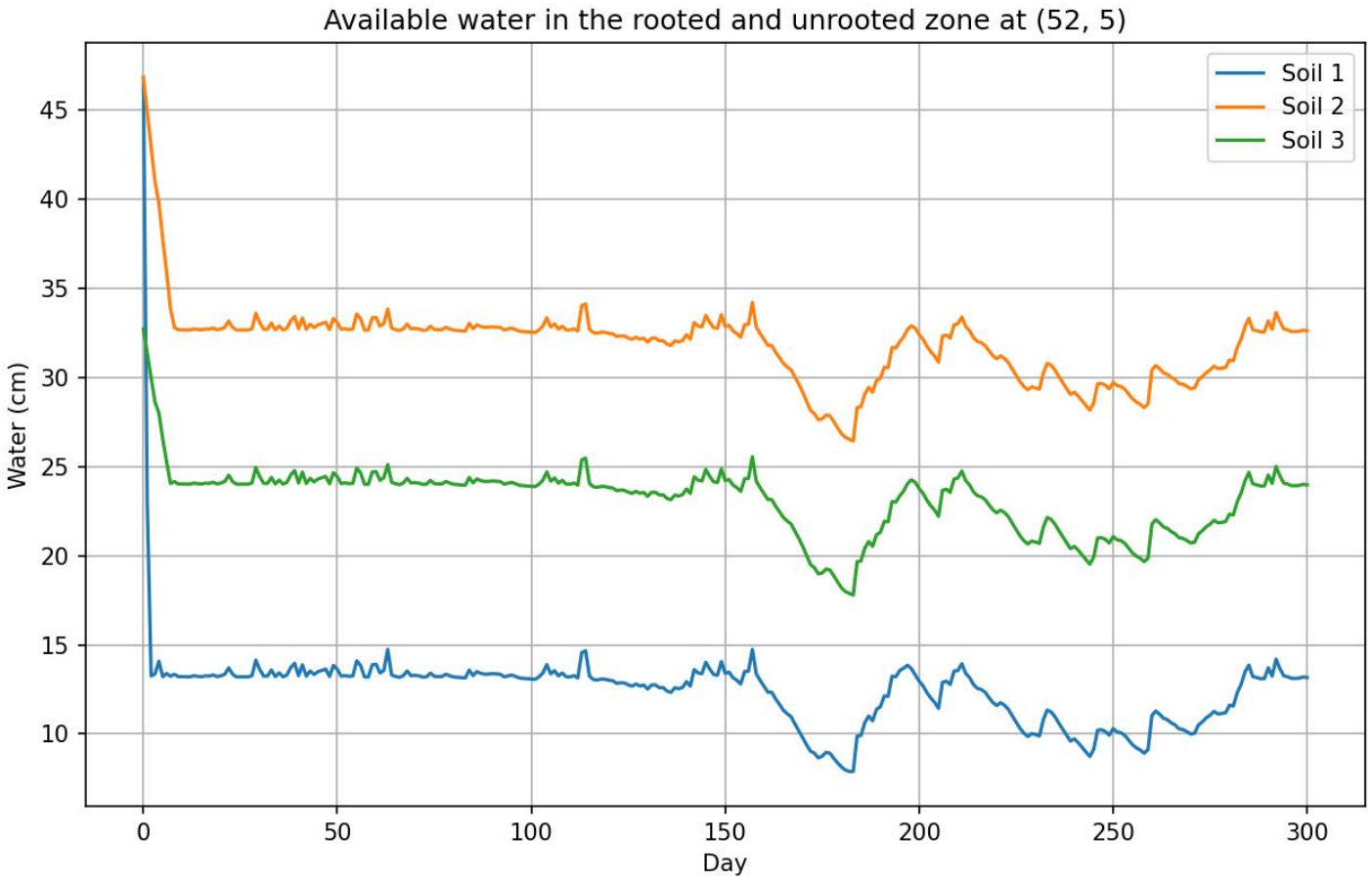
Batch Simulation for Synthetic Data

```
def run_virtual_field(lat, lon, cropfile, soilfile, agrofile):
    weather = NASAPowerWeatherDataProvider(latitude=lat, longitu
    cropdata = CABOFileReader(cropfile)
    sitedata = WOFOST72SiteDataProvider(WAV=100)
    soildata = CABOFileReader(soilfile)
    parameters = ParameterProvider(cropdata=cropdata, soildata=s
    agro = YAMLAgroManagementReader(agrofile)
    wofost = Wofost72_WLP_CWB(parameters, weather, agro)
    wofost.run_till_terminate()
    return wofost.get_output()
```

Examples of outputs

Given three different soil moisture characteristics, i.e., porosity, saturation level, and density, ..., we can generate different simulation for a same region, same crop, same weather conditions. Here are some example.

Examples of outputs



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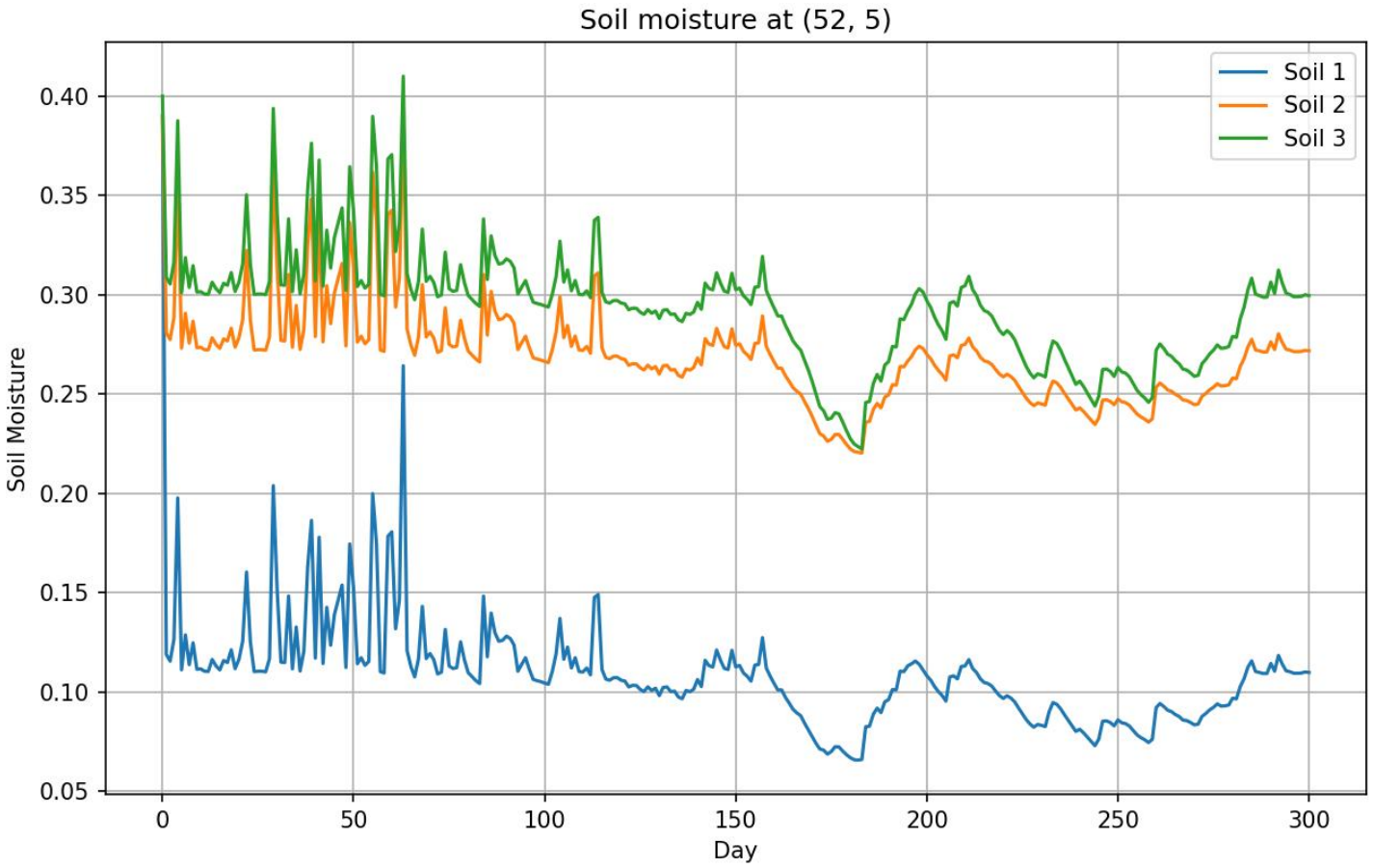

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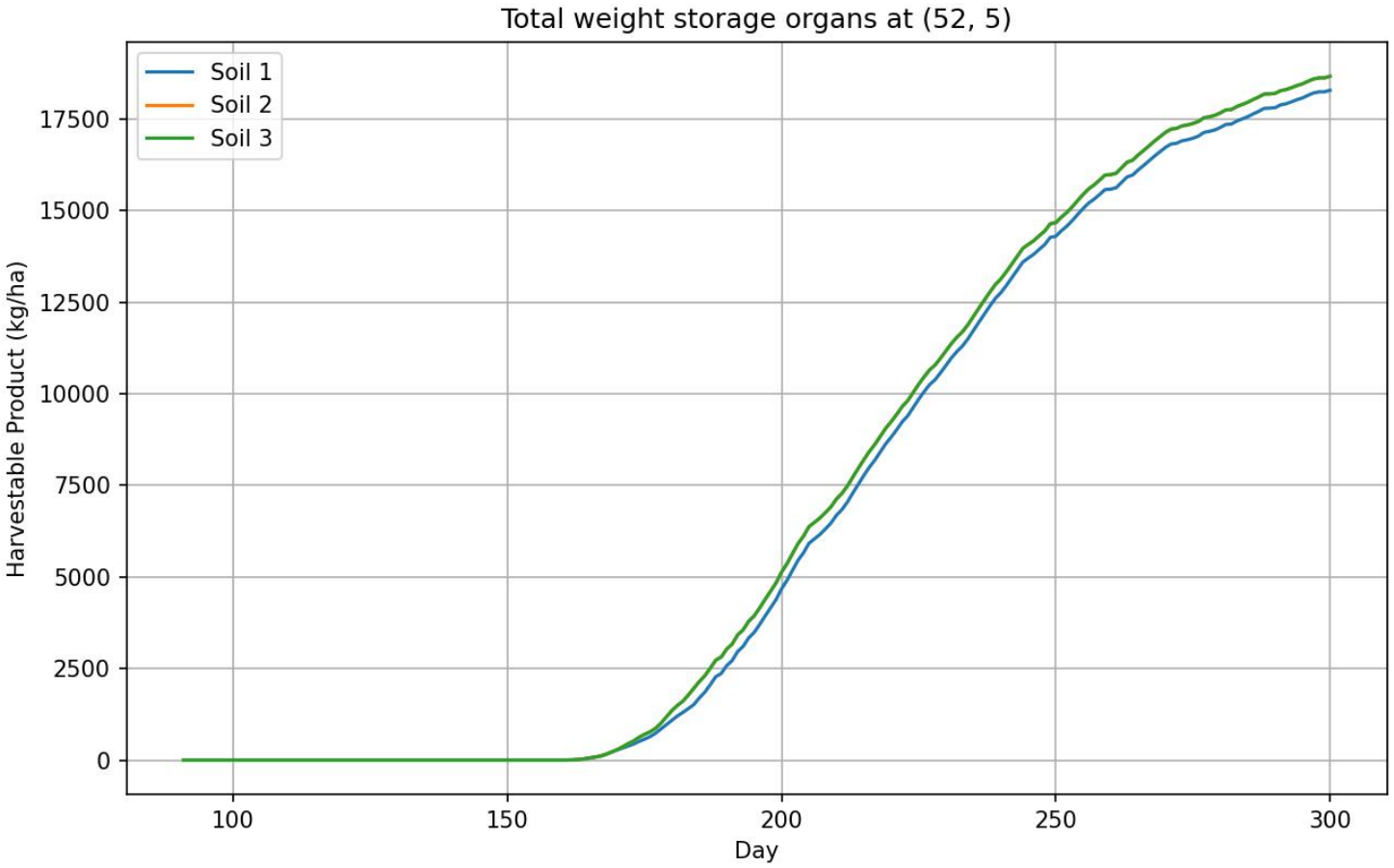

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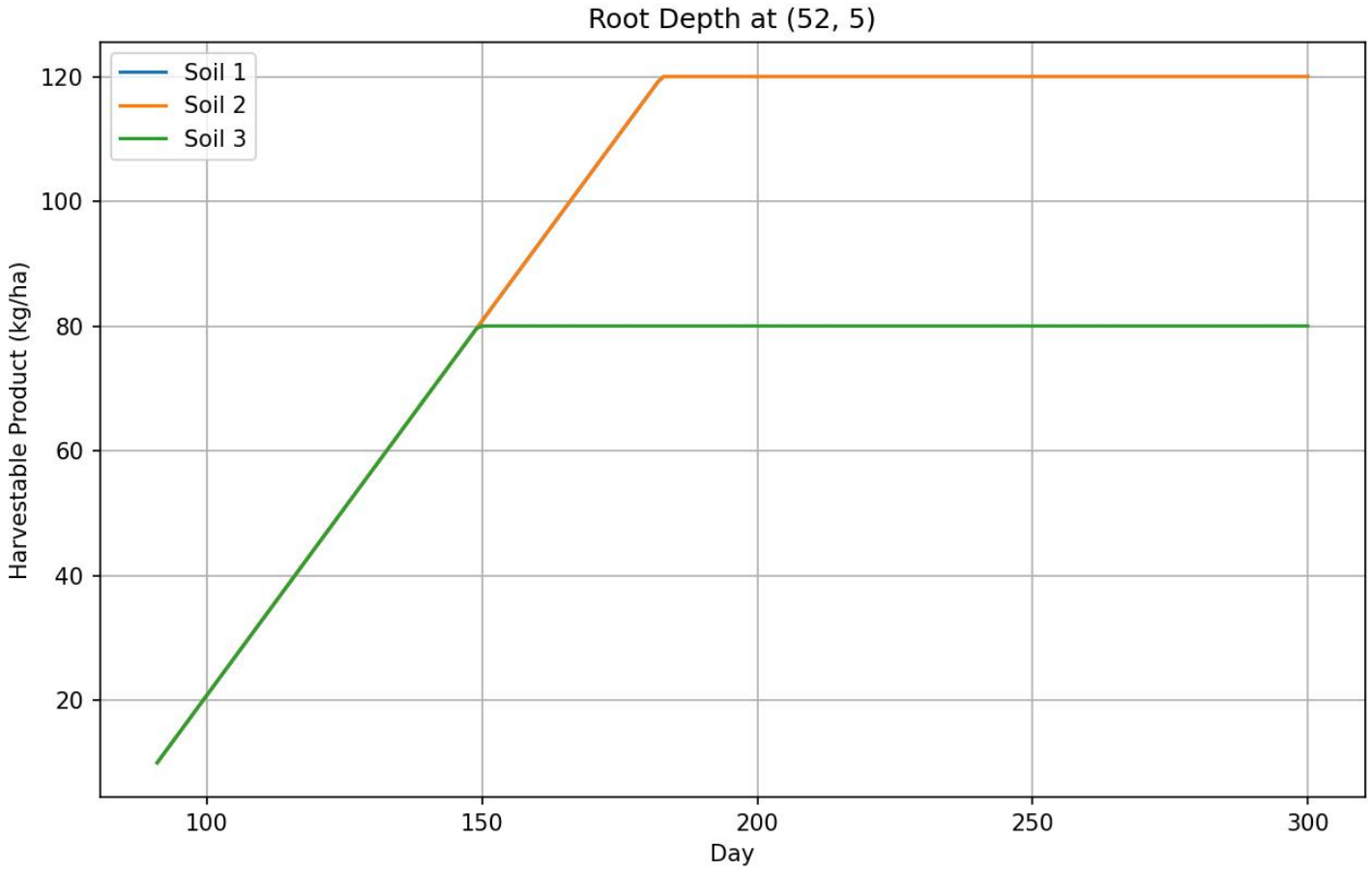

PCSE within a Digital Twin

- **Mechanistic model:** PCSE simulates the biophysical dynamics.
- **Data layer:** weather stations, soil moisture sensors, satellite indices.

Roles of PCSE:

- Mechanistic backbone of the twin.
- Synthetic data generator for training and benchmarking.
- Can also serve as a reference model for surrogate or hybrid learning.

Hybrid ML-Physics Opportunities

- **Surrogate modeling:** learn differentiable emulators of PCSE dynamics.
- **Hybrid modeling:** replace empirical submodules (e.g., photosynthesis) by learned ones.
- **Data assimilation:** align another simulation with PCSE synthetic observations using EnKF or Bayesian learning.
- **Causal analysis:** use PCSE as a controlled environment for sensitivity or explainability studies.

Key Takeaways

- PCSE: domain-specific, ODE-based simulator for crop-soil-climate systems.
- Ideal as a **data generator** and **mechanistic core** for agricultural digital twins.
- Enables hybrid workflows combining physics, data, and control.
- Fully in Python, reproducible, extensible, and compatible with ML pipelines.